

Tanzim Hossain Safin

AI SECURITY & PRIVACY RESEARCHER

Dhaka, Bangladesh · tanzimhossainsafin756@gmail.com · +880 1752 329950

github.com/TanzimHossainSafin · [linkedin.com/in/tanzim-hossain-safin](https://www.linkedin.com/in/tanzim-hossain-safin) · [Google Scholar](https://scholar.google.com/citations?user=...) · [ORCID: 0009-0005-2784-5630](https://orcid.org/0009-0005-2784-5630)

RESEARCH INTERESTS

Security and privacy of retrieval-augmented and multi-agent LLM systems; adversarial evaluation of distributed AI pipelines, including data poisoning, membership inference, and knowledge extraction; differential privacy; trustworthy AI for high-stakes domains such as healthcare.

EDUCATION

BRAC University, Merul Badda, Dhaka, Bangladesh

February 2022 – August 2026

Bachelor of Science in Computer Science and Engineering (CSE)

Cumulative GPA: 3.63/4.00

Thesis: *A Security Evaluation Framework for Distributed Retrieval-Augmented Generation Systems*

PUBLICATIONS

1. **Tanzim Hossain Safin**, Sharif Noor Zisad, Ragib Hasan, and Swakkhar Shatabda. “Trust Propagation and Structural Containment in Multi-Agent LLM Pipelines.” *6th IEEE Cyber Awareness and Research Symposium (CARS 2026)*, University of North Dakota, Grand Forks, ND, USA, October 2026. **Accepted**; camera-ready submitted. Proceedings published on IEEE Xplore and indexed in Scopus.

RESEARCH EXPERIENCE

Undergraduate Thesis — Security Evaluation of Distributed RAG Systems

May 2025 – August 2026

BRAC University · Advisor: Dr. Swakkhar Shatabda

- Developed a seven-step, architecture-agnostic security evaluation framework for distributed RAG systems, deriving six attack classes from their unique distributed components: Data Poisoning, Source Selection Manipulation, Knowledge Extraction, Membership Inference, Selective Forwarding, and Denial of Service.
- Evaluated the framework on two structurally different systems, DRAG and Reliable-dRAG, revealing significant vulnerabilities: knowledge extraction recovered up to 42% of documents, membership inference reached AUC-ROC 0.90, and poisoning reduced DRAG F1 by up to 24 points while achieving 100% attack success on Reliable-dRAG.
- Uncovered Grounding-Farming, a novel attack that manipulates blockchain-based reliability scores while bypassing the system’s existing defense.
- Designed and independently evaluated six targeted defenses for DRAG, restoring poisoning-degraded F1 to 80–84%, reducing extraction to $\leq 0.02\%$, lowering membership-inference AUC-ROC from 0.88 to 0.76, and reducing dropped queries by up to 74.6%.
- Swept defenses across attack intensities, finding that three defenses became stronger as attacks intensified, while the congestion-based DoS defense weakened under heavy attack load.
- *Implementation*: Python, Sentence-BERT embeddings, scikit-learn, self-hosted Ollama inference.

Trust Propagation and Structural Containment in Multi-Agent LLM Pipelines

April 2026 – October 2026

Peer-reviewed and accepted at IEEE CARS 2026 (see Publications)

- **Threat model**: In multi-agent LLM systems, privilege is inherited through natural-language context rather than enforced structurally. A compromised low-privilege agent can therefore forge authority in the text it passes downstream and induce a high-privilege agent to execute a real, unauthorized action.
- **System**: Built a four-agent LangGraph pipeline (Supervisor, Researcher, Validator, Executor) with an instrumented Observer layer recording every inter-agent message, so attack propagation could be traced hop by hop rather than inferred from the final output.
- **Attacks**: Evaluated two realistic attack vectors, memory poisoning and indirect prompt injection, against the Validator’s semantic judgment, confirming that a purely LLM-based reviewer can be reliably deceived by adversarial context.
- **Defense**: Implemented an authorization layer independent of agent judgment, pairing cryptographically signed capability tokens with a policy oracle checked at execution time, so that the Executor derives permission from verifiable structure rather than from upstream claims.
- **Results**: The structural authorization layer reduced unsafe execution from 100% to 0% across both attacks while leaving legitimate task completion unaffected, demonstrating that containment does not require solving adversarial robustness at the reasoning layer.
- **Metric**: Introduced the Judgment Bypass Rate, which separates a deceived reviewer from an actual downstream safety failure. Conventional metrics conflate the two and overstate risk where a deceived agent is still structurally blocked from acting.

PROJECTS

AI-Powered Financial System

October 2025 – December 2025

- **Problem**: Individuals managing personal finances often lack a single, intelligent tool to track spending, assess risk, and plan investments, leading to fragmented budgets and poor financial literacy.
- **Solution**: Built a full-stack AI-powered financial platform (React/Tailwind frontend, Node.js/Express backend) for budgeting, investment tracking, and expense optimization, using intelligent recommendations to help users understand spending habits and plan investments.
- **Features**: Automated budget organization, spending-habit analysis, investment planning, and risk assessment, with AI-driven recommendations surfaced through a REST API and a React-based interface.

TECHNICAL SKILLS

- **Languages**: Python, TypeScript, JavaScript, C, C++
- **ML & Research Tooling**: PyTorch, Scikit-learn, Hugging Face Transformers, Sentence-BERT, LangGraph, Ollama, NumPy, Pandas, Matplotlib
- **Infrastructure & Tools**: Git, Docker, Linux, Kaggle/Colab GPU training
- **Web & Databases**: Node.js, Express, React, Next.js, PostgreSQL, MongoDB, MySQL